

So

$$\|\partial_I \Phi(n, I) \cdot h\|_\infty \geq \|h\|_\infty. \quad (43)$$

ineq:diff

Therefore, $\partial_I \Phi(n, I)$ is a global diffeomorphism on $\mathcal{C}^0([0, T], \mathbb{R})$ and according to (43)

$$\left\| \left(\partial_I \Phi(n, I) \right)^{-1} \right\| \leq 1.$$

Hence, according to the implicit function Theorem, we can find a function $F \in \mathcal{C}^1(X, \mathcal{C}^0([0, T], \mathbb{R}^+))$ such that

$$\forall n \in X, \quad \Phi(n, F(n)) = 0.$$

i.e.

$$\forall t \in [0, T], F(n)(t) = \int_0^\infty r(F(n)(t), x) n(t, x) dx. \quad (44)$$

eq:fonction

Finally

$$\|\partial_n F(n)\| \leq \left\| \left(\partial_I \Phi(n, I) \right)^{-1} \right\| \|\partial_n \Phi(n, I)\| \leq 1 \times r_M.$$

Hence, F is r_M -lipschitzian on X .

Now, if μ_1, μ_2 are two solutions of (26), associated to the activity function $I_1, I_2 \in \mathcal{C}^0([0, T], \mathbb{R})$ with $\mu_1(0) = \mu_2(0) = n_{\text{ini}}$, then, according to Lemma 6.2,

$$\forall t \in [0, T], \quad \|\mu_1(t) - \mu_2(t)\|_1 \leq \|\mu_1(0) - \mu_2(0)\|_1 + 2\|\partial_I r\|_\infty \int_0^t \|I_1(\tau) - I_2(\tau)\|_\infty d\tau$$

Then, since $\mu_1(0) = \mu_2(0) = n_{\text{ini}}$, we obtain

$$\forall t \in [0, T], \quad \|\mu_1(t) - \mu_2(t)\|_1 \leq 2T\|\partial_I r\|_\infty \|I_1 - I_2\|_\infty.$$

And so

$$\|\mu_1 - \mu_2\|_X \leq 2T\|\partial_I r\|_\infty \|I_1 - I_2\|_\infty.$$

Finally, if $n_1, n_2 \in X$, and $\mu(n_1), \mu(n_2) \in X$ are the two solutions associated to the activity functions $I_1 = F(n_1)$ and $I_2 = F(n_2)$ and with $\mu_1(0) = \mu_2(0) = n_{\text{ini}}$, then,

$$\begin{aligned} \|\mu(n_1) - \mu(n_2)\|_X &\leq 2T\|\partial_I r\|_\infty \|I_1 - I_2\|_\infty \\ &\leq 2T\|\partial_I r\|_\infty r_M \|n_1 - n_2\|_X \end{aligned}$$

Now, if we take $T > 0$ such that $2T\|\partial_I r\|_\infty r_M < 1$, then the function $n \mapsto \mu(n)$ is a contraction on X and therefore, according to Picard Fixed point Theorem, there exist a unique function on X such that $\mu(n) = n$. This function is the solution of (1) on $[0, T]$.

Finally, we use the same argument on every interval $[kT, (k+1)T]$, with $k \in \mathbb{N}$ to prove that the solution n is uniquely and well-defined on $\mathbb{R}^+$. $\square$

rapport a r

Lemma 6.2. *Let μ_1 and μ_2 be the solution of (26) with respectively $r_1(t, x) := r(x, I_1(t))$ and $r_2(t, x) = r(x, I_2(t))$, $\mu_1(0) = \mu_1^0$ and $\mu_2(0) = \mu_2^0$. Then,*

$$\forall t \geq 0, \|\mu_1(t) - \mu_2(t)\|_1 \leq \|\mu_1^0 - \mu_2^0\|_1 + 2\|\partial_I r\|_\infty \int_0^t \|I_1(\tau) - I_2(\tau)\|_\infty d\tau.$$